# RUNCHEN HOU

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## EDUCATION

**University of Illinois at Urbana-Champaign** Expected May 2027
*Bachelor of Science in Computer Engineering* GPA: 3.85/4.00
**Related Coursework:**
Compilers and OS MySQL Database System Data Structure & Algorithm Circuit Design & Validation

## TECHNICAL SKILLS

**Programming & Tools:** C/C++, Python, Bash/Shell, Git, Linux (Ubuntu/CLI), RISC-V Assembly, Swift
**Cloud Computing:** React, REST API, Node.JS, Google Cloud, MySQL/NoSQL, Orcas Agent Platform, Agent Development Kit

## WORK EXPERIENCE

**Google** San Diego, California
*Silicon Engineering Intern* May 2026 – August 2026
- Delivered an internal agentic phone testing robot by leading the software and hardware teams
- Enabled agentic CI/CD integration to Orcas Agent Platform with client CLI and a web dashboard
- Eliminated need for human intervention in chip validation process by enabling remote access
- Easy tracking through an SQLite database with triple-threaded Flask/Waitress engine and REST APIs
- Flashed ESP32 C++ firmware to control Z-actuators and servos via JSON protocol over TCP/Serial

**University of Illinois, Computer Systems & Engineering**
*Teaching Assistant* September 2024 – May 2025
- Sped up grading by two hours per week by inventing a Bash-based smart Git Autograder

**JoeyLearn**
Startup Founder, Cozad New Venture Challenge October 2026 – Present
- Revolutionized pre-college education with AI educational gameplay with central teacher controls & integration

## PUBLICATION & PROJECT HIGHLIGHTS

**Improved Google AI Mode Markdown Parsing** San Diego, California
Owner August 2026 – Present
- Eliminated formatting faults present in Google AI Mode in Google App by improving Streamdown parser
- Improved parsing with speculation when streaming partial LLM output involving bold and italics

**Towards Proactive Personalization through Profile Customization for Individual Users in Dialogues** Haining, China
*AI Research Co-Author, ACL 2026 Findings* April 2025 – May 2026
- Devised a Python Machine Learning personalization algorithm to actively remember user preferences
- 20 percent faster responses and cheaper cost by reducing memory overhead and model reaction time

**Arxivgram: Academic Paper Social Network** Champaign, Illinois
*NodeJS and MySQL Head Engineer* September 2025 – December 2025
- Led full-stack development of a social network with a NodeJS frontend and MySQL backend on Google Cloud
- Developed a Python AI paper recommendation algorithm with a 50,000+ paper JSON dataset

## EXTRACURRICULAR ACTIVITIES & LEADERSHIP

**UIUC Retrocomputing Club** Champaign, Illinois
Contributor August 2025 – Present
**Varsity Volleyball Team** Haining, China
*Outside Hitter* September 2023 – Present
**FIRST Tech Challenge Robotics Team** Chengdu, China
*Senior Software Engineer* September 2020 – May 2023
- Programmed a Python-based Pytorch system for AI “donut ring” object detection
- Delivered a belt system for lightning-fast ring ejection mechanism with aiming assistance